CV Quiz-1 (version-2)

Duration: 1 hour — Total Marks: 40 marks — Single-correct MCQs

Correct ans: +1 marks **Wrong ans: -1/4 marks**

1. "Circle of confusion" size increases when:

(A) Object is at exact focus distance
(B) Aperture is smaller (all else fixed)
(C) Object is away from focus and aperture is larger
(D) Sampling rate increases

Answer: C

2. To increase depth of field while keeping exposure roughly constant, you generally need to:

(A) Close aperture and decrease exposure time
(B) Close aperture and increase exposure time/ISO
(C) Open aperture and increase exposure time
(D) Open aperture and decrease exposure time

Answer: B

3. Under the thin-lens model (relation not shown), with $f = 35$ mm and object distance 700 mm, image distance is closest to:

(A) 33.3 mm
(B) 35.0 mm
(C) 36.8 mm
(D) 70.0 mm

Answer: C

4. Changing aperture from f/4 to f/2.8 with all else fixed changes light by approximately:

(A) Half
(B) No change
(C) Double
(D) Four times

Answer: C

5. Under ideal perspective projection of a rigid 3D scene to a 2D image, which property is *guaranteed* to be preserved?

(A) Euclidean lengths of segments
(B) Angles between intersecting lines
(C) Collinearity of points (straightness)
(D) Parallelism of 3D lines in general

Answer: C

6. A 2D point is represented in homogeneous coordinates as $\tilde{p} = [120,\ 80,\ 40]^T$. Its Cartesian coordinates (x, y) are:

(A) $(120, 80)$
(B) $(3, 2)$
(C) $(2, 3)$
(D) $(40, 40)$

Answer: B

7. A pinhole camera has its center of projection at the origin and an image plane at $z = f$. Under the standard perspective mapping, for which 3D point is the projection *undefined* (i.e., cannot be assigned a finite image point)?

(A) $(3, 1, f)$
(B) $(3, 1, 0)$
(C) $(0, 0, f)$
(D) $(3, 1, 2f)$

Answer: B

8. Which statement is most correct about "brightness adjustment" vs "multiplicative gain" (contrast) in practice?

(A) Both are strictly linear operators
(B) Gain is linear; brightness shift is not strictly linear but used in practice
(C) Brightness shift is linear; gain is not
(D) Neither can be analyzed mathematically

Answer: B

9. An 8-bit image has intensities: 10 in 40% pixels, 200 in 60%. After standard histogram equalization (CDF scaled to 0–255 and floored), the mapped value of intensity 10 is:

(A) 51
(B) 102
(C) 153
(D) 204

Answer: B

10. In an 8-bit image, you apply Brightness shift by -50 with clipping to [0,255]. Pixel value 20 becomes:

(A) -30
(B) 0
(C) 20
(D) 50

Answer: B

11. Gamma correction with $\gamma = 0.5$ (sqrt-like), normalized 0–1 then rescaled 0–255. Input intensity is 64. Output is closest to:

(A) 32
(B) 64
(C) 96
(D) 128

Answer: D

12. Unsharp masking is closest to adding back:

(A) A scaled residual (original − blurred)
(B) A scaled blurred image only
(C) A median-filtered image
(D) Histogram CDF

Answer: A

13. Consider the kernel

$$K = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

Which statement is correct?

(A) K is not separable
(B) K is separable
(C) K is separable only after normalization to sum 1
(D) K is separable only if used with circular padding

Answer: B

14. A 3×3 box filter is applied to a 3×3 patch with all pixels 90. Center output is:

(A) 10
(B) 30
(C) 90
(D) 270

Answer: C

15. A $k \times k$ Gaussian kernel G_σ is separable into $g_x \cdot g_y^T$. The computational saving over direct convolution for an $N \times N$ image is:

(A) $O(N^2k^2) \to O(N^2k)$
(B) $O(N^2k) \to O(N^2 \log k)$
(C) $O(k^2) \to O(\log k)$
(D) $O(N^2k^2) \to O(N^2 \log k)$

Answer: A

16. If a filter h is anti-symmetric, then $I * h$ is equivalent to:

(A) Correlation with h
(B) Correlation with $-h$
(C) Smoothing
(D) Identity

Answer: B

17. In Unsharp masking, for $I = 80$, $I_{\text{blur}} = 70$, $\gamma = 2$, output is:

(A) 90
(B) 95
(C) 100
(D) 110

Answer: C

18. You replace convolution by correlation in a pipeline and the output remains unchanged on all test images. Which must be true about the kernel?

(A) Kernel has zero mean
(B) Kernel is even-symmetric about its center
(C) Kernel is separable
(D) Kernel has odd size

Answer: B

19. Treating interpolation as convolution implies that upsampling ideally should be preceded/followed by:

(A) High-pass only
(B) Appropriate low-pass reconstruction to suppress imaging artifacts
(C) Histogram equalization
(D) Phase randomization

Answer: B

20. Bilinear interpolation on 2×2 block: TopLeft=0, TopRight=40, BottomLeft=20, BottomRight=60. At $(x, y) = (0.5, 0.5)$, interpolated value is:

(A) 20
(B) 30
(C) 40
(D) 60

Answer: B

21. Nearest-neighbor interpolation in 1D: values at positions 0 and 1 are 0 and 100. Query at 0.49 gives:

(A) 0
(B) 49
(C) 51
(D) 100

Answer: A

22. Aliasing is prevented by:

(A) High-pass before sampling
(B) Low-pass before sampling to satisfy Nyquist
(C) Histogram equalization before sampling
(D) Phase unwrapping

Answer: B

23. Downsampling an image without anti-aliasing mostly corrupts:

(A) Low frequencies
(B) High frequencies folding into low frequencies
(C) DC component only
(D) Histogram CDF

Answer: B

24. A signal contains frequencies up to 6 kHz. Minimum sampling rate to avoid aliasing is:

(A) 6 kHz
(B) 9 kHz
(C) 12 kHz
(D) 24 kHz

Answer: C

25. An ideal high-pass filter that removes the lowest 5% of frequencies (near DC) most directly causes:

(A) Global illumination/slow variations removed
(B) Strong blur
(C) Uniform histogram
(D) Reduced aliasing

Answer: A

26. Downsampling by 2 without anti-aliasing primarily causes:

(A) Only DC to change
(B) High-frequency folding into low frequencies
(C) Uniform histogram
(D) Removal of phase information

Answer: B

27. Hybrid images are constructed by combining:

(A) Low-frequency components of two images
(B) High-frequency components of two images
(C) Low-frequency components of one image with high-frequency components of another image
(D) Phase of one image with magnitude of another image

Answer: C

28. In a hybrid image, the image perceived from a far viewing distance is dominated by:

(A) High-frequency components
(B) Low-frequency components
(C) Phase information
(D) Edge orientations

Answer: B

29. Why does the human visual system perceive different images at different viewing distances in hybrid images?

(A) Because phase dominates perception at short distances
(B) Because viewing distance effectively acts as a low-pass filter
(C) Because FFT changes with scale
(D) Because color perception changes with distance

Answer: B

30. Why is careful alignment of the two source images important when constructing hybrid images?

(A) To preserve color consistency
(B) To avoid aliasing in the frequency domain
(C) To ensure corresponding semantic features overlap across frequencies
(D) To reduce FFT computation time

Answer: C

31. Which operation best explains why hybrid images fail when printed at very small sizes?

(A) Quantization noise
(B) Histogram equalization
(C) Spatial downsampling removing high frequencies
(D) Phase inversion

Answer: C

32. Which task is most directly associated with *learning-based vision*?

(A) Object recognition / detection using a trained CNN
(B) Estimating depth via triangulation (multi-view geometry)
(C) Photometric calibration and BRDF-based reflectance modeling
(D) Removing periodic stripe noise using a notch filter

Answer: A

33. (Filter response intuition) When a sinusoid of frequency ω is convolved with a linear filter $h(x)$, the output is:

(A) A sinusoid of a different frequency
(B) A sinusoid of the same frequency with changed magnitude and phase
(C) A non-sinusoidal signal
(D) Zero for all inputs

Answer: B

34. Compared to a box filter of the same spatial support, a Gaussian filter is preferred because:

(A) Its frequency response has fewer oscillations, reducing ringing artifacts
(B) It completely removes high-frequency components
(C) It preserves edges better than any other filter
(D) Its Fourier transform has compact support

Answer: A

35. For very large kernels, FFT-based convolution becomes faster than spatial convolution mainly because:

(A) FFT removes the need for kernel normalization
(B) FFT reduces convolution to pointwise multiplication in the frequency domain
(C) FFT makes convolution linear in kernel size
(D) FFT preserves only magnitude and discards phase

Answer: B

36. An image contains a sinusoidal pattern with spatial frequency 0.6 cycles/pixel. It is sampled at 1 pixel per unit length without prefiltering. What happens?

(A) The frequency is preserved exactly
(B) The frequency aliases to 0.4 cycles/pixel
(C) The frequency aliases to 0.2 cycles/pixel
(D) No aliasing occurs because images are 2D

Answer: B

37. In the Fourier magnitude visualization of a natural image, why is most energy concentrated near the center?

(A) Natural images contain more edges than smooth regions
(B) Low spatial frequencies dominate natural images
(C) High frequencies are suppressed by sensors
(D) Phase information is discarded

Answer: B

38. In a two-dimensional Fourier transform of an image, what does a frequency component with large values of both frequency coordinates primarily correspond to?

(A) Slowly varying intensity changes
(B) Uniform brightness across the image
(C) Rapid intensity changes in both spatial directions
(D) Image boundaries only

Answer: C

39. A 256×256 image is convolved with a 7×7 kernel using *valid* convolution (no padding). Output size is:

(A) 256×256
(B) 250×250
(C) 252×252
(D) 262×262

Answer: B

40. In a 2D image, a horizontal edge results in high-frequency energy along the ________ axis in the Fourier domain.

(A) Horizontal
(B) Vertical

(C) Diagonal

(D) DC only

Answer: B